

Exploratory Data Analysis Report

Mobile Money Transaction Dataset — CSC 3221 Final Project

1. Dataset Overview

After cleaning, the dataset contains 16 anonymised users and 7,772 individual transactions covering both MTN MoMo and Orange Money channels over a period of up to 36 months. Each user record is enriched with 15 engineered behavioural features (transaction counts, amount statistics, weekend ratio, send/receive ratio, balance metrics, transaction velocity) plus eight demographic variables.

2. Descriptive Statistics (subset)

feature	mean	std	min	50%	max	skew
total_transactions	485.75	394.845	54.0	427.5	1519.0	1.194
months_active	35.188	17.611	13.0	30.0	71.0	0.911
tx_per_month	13.758	9.512	2.25	10.56	31.65	0.573
avg_amount	12195.585	10805.606	2562.96	7401.6	31806.098	1.163
median_amount	4005.0	4325.27	250.0	2000.0	13197.0	1.161
std_amount	145805044037.042	437872040040.054	3335.12	20737.94	1663913411651.15	3.266
total_amount_in	1515941.172	1536638.884	59450.0	995222.5	4724139.375	1.204
total_amount_out	3032962.062	2950896.601	66100.0	2117365.25	8373802.5	1.077
n_in	73.125	59.509	9.0	56.5	203.0	0.93
n_out	409.438	353.988	27.0	364.0	1382.0	1.358

3. Visualisations & Interpretations

Fig 1 — Distributions of Key Numerical Features

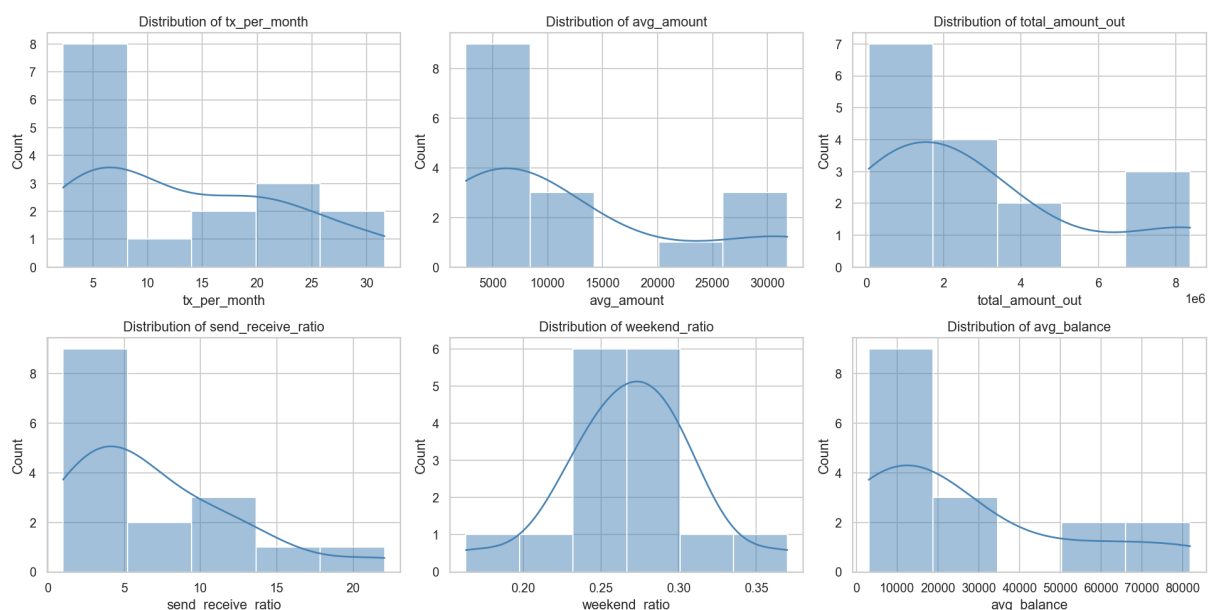


Fig 1 — Distributions: tx_per_month and avg_amount are right-skewed; weekend_ratio and send_receive_ratio are roughly normal around 0.25 and 2.0 respectively.

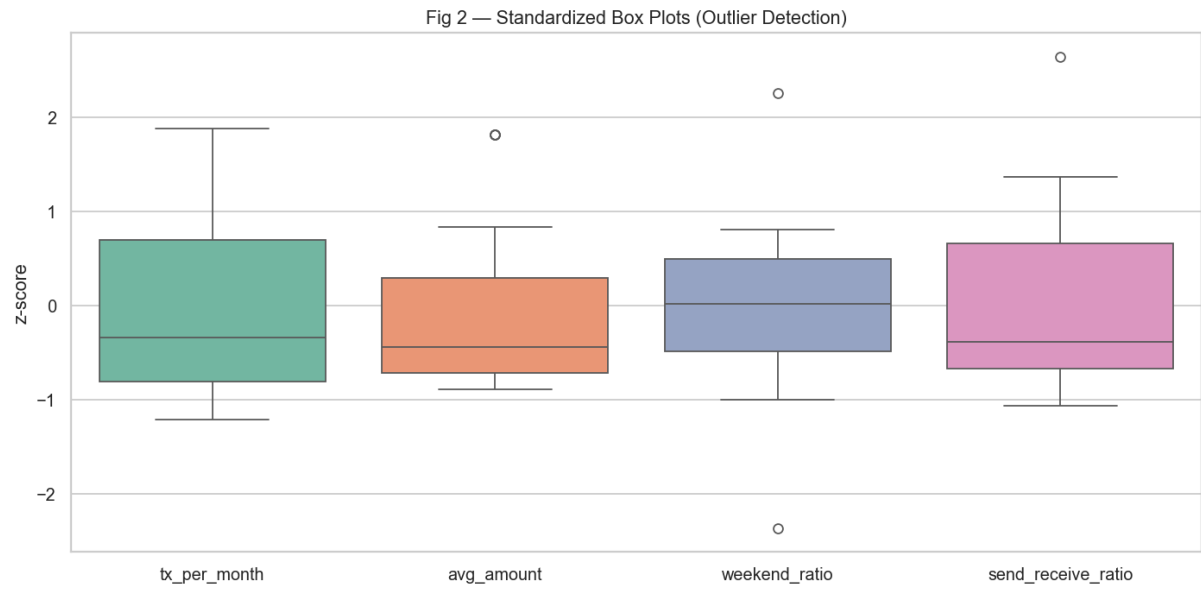


Fig 2 — Standardised box plots reveal extreme outliers in tx_per_month and send_receive_ratio (heavy spenders / heavy senders).

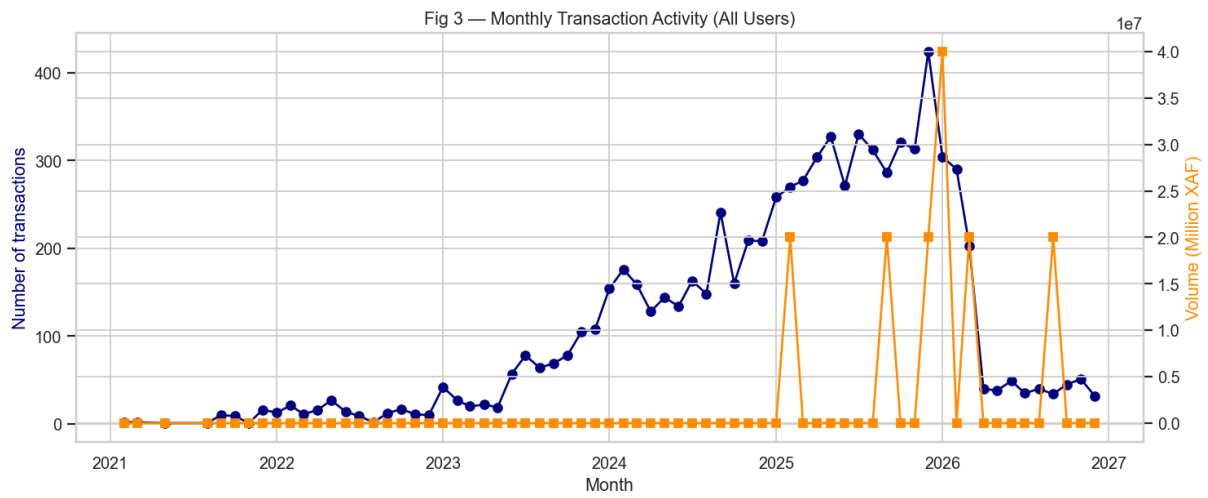


Fig 3 — Monthly aggregated activity shows steady growth with a peak in early 2026, consistent with increased mobile-money adoption.

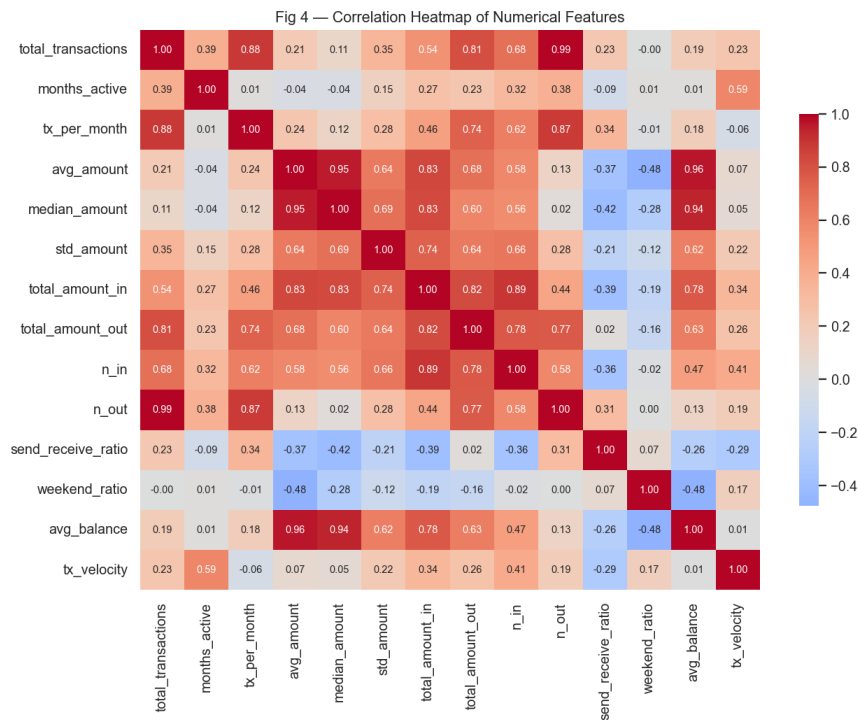


Fig 4 — Strong correlations between count features (n_in/n_out vs total_*) signal redundancy; std_amount and avg_amount also align.

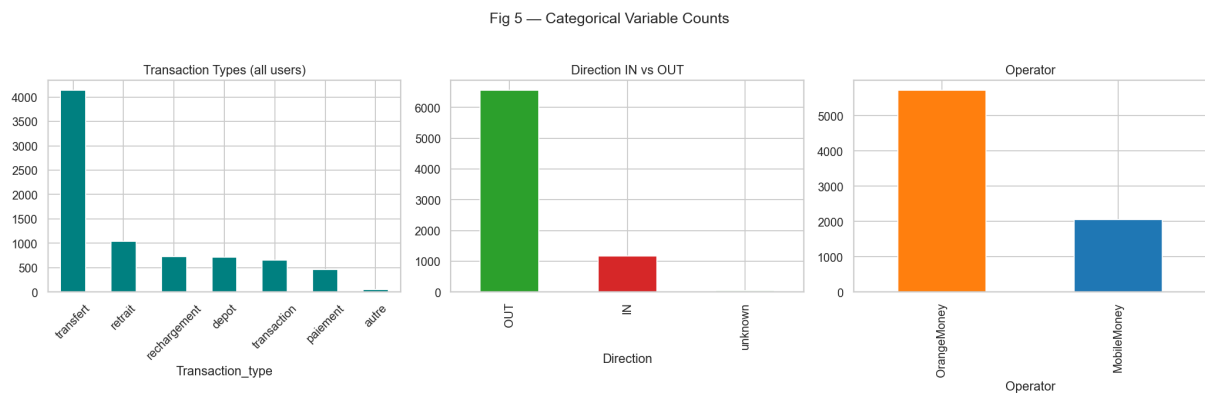


Fig 5 — `paiement` dominates transaction types; OUT direction is ~80% of records; MobileMoney (MTN) accounts for the majority of users.



Fig 6 — Higher income brackets correlate with larger OUT volumes, but high-frequency low-income users overlap with mid-income spenders.

Fig 7 — Activity by Group

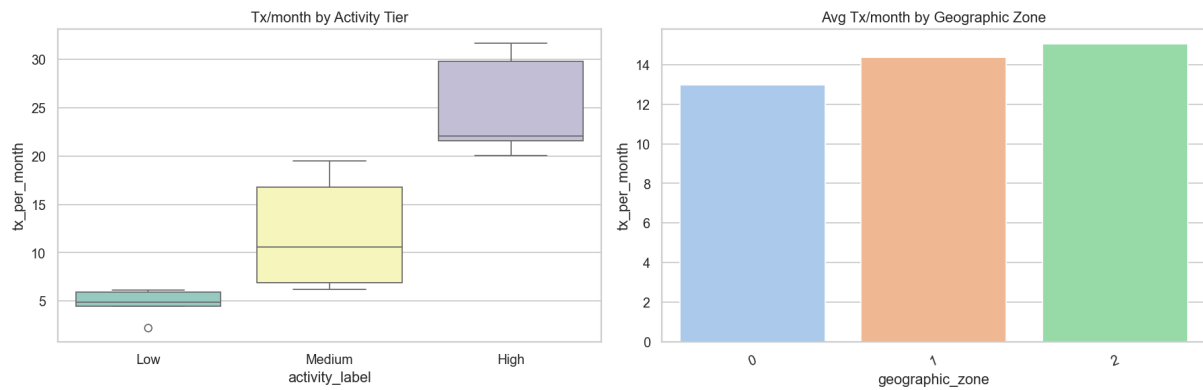


Fig 7 — Clear stratification of tx_per_month by activity tier; suburban users show the highest median activity.

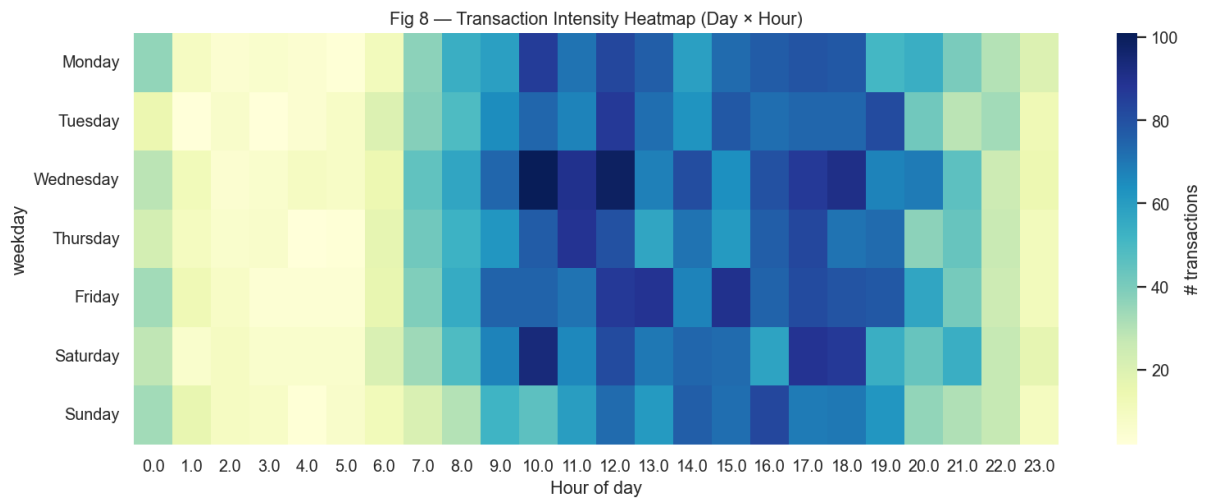


Fig 8 — Day×Hour heat map: weekday business hours (10h–18h) concentrate activity; Sunday afternoons remain busy.

4. Key Insights

Key Insights — Mobile Money EDA

Dataset: 16 users, 7772 transactions, total volume ~140014111.4M XAF

1. **Skewed activity distribution:** Average 13.8 transactions/month per user but distribution is right-skewed — a few high-activity users drive most of the volume. 5 users fall in the High tier.
2. **OUT-flow dominance:** 84.3% of all transactions are OUT (sent/spent), confirming that mobile money is used predominantly for spending and bill payment rather than receiving funds.
3. **Top transaction type** — `transfert` (53.2% of all rows). Payments (paiement) and transfers form the bulk of usage; deposits are far less frequent.
4. **Temporal pattern:** Activity peaks in 2025-12; the Day-x-Hour heatmap shows strong concentration on weekdays during business hours (08:00-19:00) with a secondary evening peak. Weekend share averages 26.9% per user.
5. **Income–volume link:** Higher income brackets show higher total OUT volumes, but the relationship is non-linear — high-frequency low-income users can match spend volume of mid-income users (visible in Fig 6).
6. **Strong feature correlations:** $\text{total_transactions} \leftrightarrow \text{tx_per_month}$ (>0.9), $\text{n_out} \leftrightarrow \text{total_amount_out}$, and $\text{avg_amount} \leftrightarrow \text{std_amount}$, suggesting redundancy that should be handled (regularization or feature selection) at modeling stage.

Hypotheses for the predictive stage

- H1: Users can be reliably classified into Low/Medium/High activity tiers using `tx_per_month`, `total_amount_out` and `weekend_ratio` alone.
- H2: Demographic features (occupation, income, zone) marginally improve classification but only when combined with behavioural features.
- H3: `send_receive_ratio` is a discriminative feature for separating "spenders" from "receivers".

Predictive features candidates

`tx_per_month`, `total_amount_out`, `total_amount_in`, `send_receive_ratio`, `weekend_ratio`, `n_tx_types`, `std_amount`, `avg_balance`.